Computer Organization and Architecture
(Code: CS256)

Lecture-16: Floating Numbers

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Real or Fraction Number

- *Real number is a quantity that can be expressed as an infinite decimal expansion.*
- Real numbers are used in measurements of continuously varying quantities such as size and time.
- In contrast, the natural numbers 1, 2, 3, . . . , arising from counting.
- *A real also known as fraction number, is a whole number with a decimal point.*
- For example, 5.5, 0.25, and -103.342 are all real numbers, while 91, and 0 are not.
- *Real numbers can be defined as the union of both rational and irrational numbers.*
- They can be both positive or negative.
- They are denoted by the symbol "R".

Real Number Classifications by Representation

- There are two major approaches to store real numbers (i.e., numbers with fractional component) in modern computing.
- A fraction number can be seen in
  - Fixed Point Representation
  - Floating Point Representation
- In fixed point notation, there are a fixed number of digits after the decimal point.
  - Example: 25.80, 25.00, 27.89, etc.
- Floating point number allows for a varying number of digits after the decimal point.
  - Example: 25.80, 25.001, 27.8966, etc.

Fixed Point Representation

- This representation has fixed number of bits for integer part (I) and for fractional part (F).
- For example, if given fixed-point representation is IIII.FFFF, then you can store minimum value is 0000.0001 and maximum value is 9999.9999.
- There are three parts of a fixed-point number representation:
  - sign field,
  - integer field, and
  - fractional field.

Sign Bit	Integer Part	Fractional Part

Signed Fixed point representation

Integer Part	Fractional Part

Fixed Point Advantage and Disadvantage

- The advantage of using a fixed-point representation is performance.
- Main disadvantage is relatively limited range of values that they can represent.
- *So, it is usually inadequate for numerical analysis as it does not allow enough numbers and accuracy.*
- *A number whose representation exceeds 32 bits would have to be stored inexactly.*
- *It is important to note that real numbers suffer from loss of precision when represented with a fixed number of bits* (e.g., 32-bit or 64-bit).

Fixed Point Advantage and Disadvantage

- This is because there are infinite number of real numbers (even within a small range of says 0.0 to 0.1).
- On the other hand, a n-bit binary pattern can represent a finite 2^n distinct numbers.
- Hence, not all the real numbers can be represented.
- The nearest approximation will be used instead, resulted in *loss of*
- Also, floating number arithmetic is very much less efficient tha arithmetic.
- *It could be speed up with a so-called dedicated floating-point co-p*
- Hence, use integers if application does not require floating-

Floating Point Number

- Real number are also known as Floating point numbers due to varying position of the decimal point.
- We can move the radix point either left or right.
- Floating point numbers get their name from the way the decimal point can "float" to any position necessary.
- Due to this, floating point numbers are often referred to as floats.

Why Floating Point?

- *To an engineer* building a highway, it does not matter whether it's 10 meters or 10.0001 meters wide - their measurements are probably not that accurate in the first place.
- *To someone designing* a microchip, 0.0001 meters (a tenth of a millimeter) is a huge difference - But they'll never have to deal with a distance larger than 0.1 meters.
- To satisfy the engineer and the chip designer, a number format has to provide accuracy for numbers at very different magnitudes.
- However, only relative accuracy is needed.
- Basically, having a fixed number of integer and fractional digits i not useful - and the solution is a format with a floating point.

Structure of Floating Point Number

- The structure of a floating point(real) number is as follows:
- $4.2 * 10^8$

Exponent

Mantissa

Base

- Only the mantissa and the exponent are stored.
- The base is implied (known already)
- As it is not stored this will save memory capacity

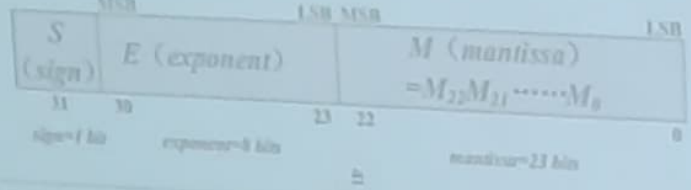
Floating-Point Decimal Number

$$
\begin{aligned}
123456.\times 10^{-1} &= 12345.6\times 10^{0} \\
&= 1234.56\times 10^{1} \\
&= 123.456\times 10^{2} \\
&= 12.3456\times 10^{3} \\
&= 1.23456\times 10^{4} \text{ (normalised)} \\
&\approx 0.12345\times 10^{5} \\
&\approx 0.01234\times 10^{6}
\end{aligned}
$$

12345

IEEE 754 for FP Numbers: Classification

- According to IEEE 754 standard, the floating-point number is represented in **four** ways: 16, 32, 64 and 128 bit.
- Sometimes referred to as Half, Single, Double and Quadruple precision.
- *Half Precision (16 bit)*: 1 sign bit, 5 bit exponent, and 10 bit mantissa
- *Single Precision (32 bit)*: 1 sign bit, 8 bit exponent, and 23 bit mantissa
- *Double Precision (64 bit)*: 1 sign bit, 11 bit exponent, and 52 bit mantissa
- *Quadruple Precision (128 bit)*: 1 sign bit, 15 bit exponent, and 112 bit mantissa



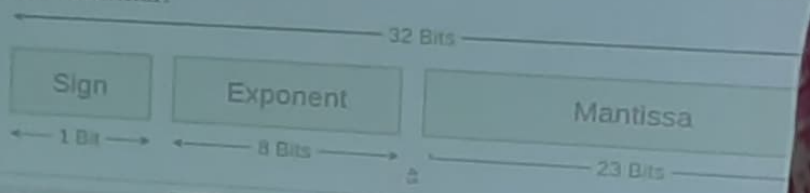
IEEE Standard for FP Numbers

- Nearly all hardware and programming languages use floating-point numbers in the same binary formats, which are defined in the IEEE 754 standard.
- Developed in response to divergence of representations
  - Portability issues for scientific code
- Now almost universally adopted

Format	Total bits	Significand bits	Exponent bits	Smallest number	Largest numb
Single precision	32	23 + 1 sign	8	ca. $1.2 \cdot 10^{-38}$	ca. 10
Double precision	64	52 + 1 sign	11	ca. $2.2 \cdot 10^{-308}$	ca 10

IEEE 754 Single Precision

- In 32-bit single-precision floating-point representation:
- The most significant bit is the sign bit (S), with 0 for positive numbers and 1 for negative numbers.
- The following 8 bits represent exponent (E).
- The remaining 23 bits represents fraction (F).
- *The Biased exponent* – The exponent field needs to represent both positive and negative exponents. A bias is added to the actual exponent in order to get the stored exponent.
- *The Normalized Mantissa* –The mantissa is part of a number in scientific notation or a floating-point number, consisting of its significant digits. Here w have only 2 digits, i.e. O and 1. So a normalised mantissa is one with only o to the left of the decimal.



IEEE 754 Single Precision

- Let the sign bit (31) be s, the exponent (30-23) be e and the mantissa (significand or fraction) (22-0) be m.
- The valid range of the exponents is 1 to 254 (if e is treated as an unsigned number).
- The actual exponent is biased by 127 to get e i.e. the actual value of the exponent is $e - 127$.
- This gives the range: $2^{1-127} = 2^{-126}$ to $2^{254-127} = 2^{127}$.

IEEE 754 Single Precision Normalization

- The normalized significand is $1.m$ (binary dot). The binary point is before bit-22 and the 1 (one) is not present explicitly.
- The sign bit $s = 1$ for a −ve number is zero (0) for a +ve number.
- The value of a normalized number is

$$(-1)^s \times 1.m \times 2^{e-127}$$

An Example

Consider the following 32-bit pattern

1 1011 0110 011 0000 0000 0000 0000 0000

The value is

$$(-1)^{1} \times 2^{10110110-01111111} \times 1.011$$

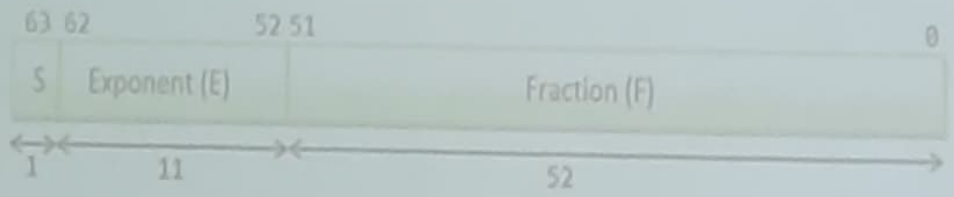
$$= -1.375 \times 2^{55}$$

$$= -49539595901075456.0$$

$$= -4.9539595901075456 \times 10^{16}$$

IEEE 754 Double Precision

- Double precision binary floating-point is a commonly used format on PCs, due to its wider range over single precision floating point, even if at a performance and bandwidth cost.
- As with single precision floating point format, it lacks precision on integer numbers when compared with an integer format of the same size.
- It is commonly known simply as double. The IEEE 754 standard defines a double as:
- Sign bit: 1 bit ; Exponent width: 11 bits ; Significand precision: 52 bits



64-bit Double-Precision Floating-point Numb

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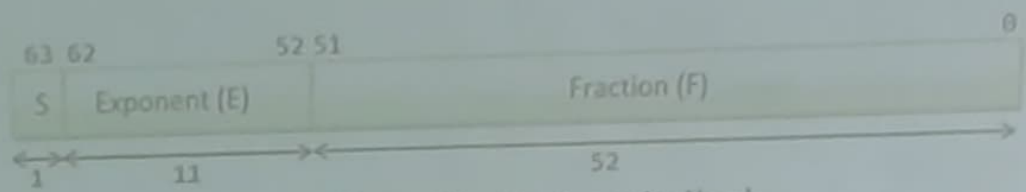

64-bit Double-Precision Floating-point Numb